AI and the Environment

SMaiLE Project

Key Information

Target Group: 4 - 7 y.o.

Duration: 3 Weeks (30–40 minute sessions)

Key Learning Goals:

1. **Scientific Understanding:** Recognize that using electricity and technology affects the planet.
2. **Digital Awareness:** Identify simple energy-saving actions such as turning off devices.
3. **Critical Thinking:** Suggest ways robots and AI can help the environment.
4. **Creativity:** Build or draw an eco-city showing nature and technology together.

Learning Outcomes

By the end of the workshop, children will be able to:

KNOWLEDGE & UNDERSTANDING:

- Understand that energy comes from resources like water, sun, and wind.
- Recognize that computers and robots need energy to work.

SKILLS & ABILITIES:

- Work in pairs or teams to play games and count energy points.
- Share ideas through drawing, storytelling, or role-play.
- Build models with Lego, paper, or Minecraft to express eco-friendly ideas.

ATTITUDES & VALUES:

- Show care for the environment through small daily actions.
- Appreciate how innovation can help build a greener future.

European Dimension / Erasmus+ Connection

- **Digital & Ecological Citizenship:** Introduces responsibility with digital tools early on.
- **Transnational Solutions:** Compares energy-saving actions across homes and

countries.

- **Green Deal Objectives:** Emphasizes how digital innovation can support sustainability goals.

1. Resources and Tools

- **Stories and Books:** "Energy Makes Things Happen", "Planet Power", "Energy Island", "The Boy Who Harnessed the Wind", and "City Green".
- **Materials:** Energy Collector Game Cards, stickers for energy points, paper, crayons, Lego blocks, or Minecraft: Education Edition.
- **Technology:** Tablets or laptops when available.

Useful Links

- Energy Makes Things Happen
- Planet Power PDF
- Energy Island PDF
- The Boy Who Harnessed the Wind PDF
- City Green PDF
- The Water Princess PDF
- Leonora Bolt excerpt

2. Working Methods

- **Storytelling:** Using books to introduce abstract concepts like energy.
- **Gamification:** Making energy use visible through an "Energy Collector Game".
- **Creative Play:** Building eco-cities to visualize solutions.
- **Reflection:** Sharing surprises and simple energy-saving pledges for home.

Activity Overview

Week	Activity	Description
Week 1	Intro & Motivation	Storytime about energy, discussion about what uses electricity at home, and drawing energy-using objects.
Week 2	Learning Through Play	Teams play the Energy Collector Game and compare which tasks use more energy.
Week 3	Creative Application	Children build or draw an eco-city with technology helpers and green spaces.
Week 3	Reflection	Children share what surprised them and one thing they can do to save energy.

3. Introduction and Motivation

Goal: Introduce the idea that technology uses energy.

- Welcome children to a lesson about robots, computers, and energy.
- Read a story such as "Energy Makes Things Happen" or "Planet Power".
- Discuss questions like: "Does a tablet need food?" and "Where does electricity come from?"
- Invite children to draw or point to things that use energy at home.

4. Research and Learning

Goal: Understand that some actions use more energy than others.

- Divide children into two or three teams.
- Place illustrated task cards face down, for example robot vacuum, playing music, or watching video.
- Teams pick cards and collect stickers that represent energy use.
- Count the stickers together and talk about which tasks use more energy and why.

5. Creative Application

Goal: Show that technology can help the planet.

- Ask: "How can robots help nature?"
- Children work in small groups to build or draw an eco-city.
- Encourage green spaces, recycling robots, and solar panels.
- Let each group present one idea, for example "Our robot collects rubbish to help animals."

6. Reflection and Evaluation

Goal: End with practical, child-friendly action.

- Children share orally or through drawing what surprised them.
- Ask what they can do at home, such as turning off lights or unplugging a tablet.
- Encourage positive feedback for every eco-friendly idea.